PROTTOY ROY

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⋄ github.com/prottoy94

PERSONAL STATEMENT

Computer Science and Engineering undergraduate at AIUB with research experience in machine learning and two conference papers accepted in 2026. Seeking an AI/ML engineering role where I can build, evaluate, and explain models on real data.

EDUCATIONAL QUALIFICATION

Bachelor of Science in Computer Science & Engineering 2023 - 2027
American International University-Bangladesh (AIUB), CGPA: 3.99/4.00
Relevant Coursework: Machine Learning, Natural Language Processing, Algorithms, Artificial Intelligence And Expert System, Computer Vision And Pattern Recognition, Data Warehouse And Data Mining, Basic Graph Theory

Higher Secondary Certificate 2020 - 2022
Major General Mahmudul Hasan Adarsha College, GPA: 5.00 (Science)

Secondary School Certificate 2015 - 2020
Bindubasini Govt Boys High School, GPA: 5.00 (Science)

TECHNICAL SKILLS

Programming	Python, SQL, C++, Java, C#, JavaScript, PHP
Machine Learning	Time-series forecasting, NLP, computer vision, ensemble methods; CNN, LSTM, GRU, and transformer (BERT, RoBERTa) models
Generative AI	LLM APIs (OpenAI, Anthropic, Gemini), LangChain, prompt engineering
ML Libraries	TensorFlow/Keras, scikit-learn, Hugging Face Transformers, OpenCV, pandas, NumPy
Deployment	FastAPI (basic REST APIs), Docker
Tools	Git, GitHub, Jupyter, MySQL, LaTeX
Languages	English (fluent), Bengali (native)

RESEARCH PUBLICATION

Prottoy Roy, Kishor Tarafder, Md. Siam Mehedi, Ashraf Uddin, "Predicting Dengue Outbreaks Using Machine Learning and Time-Lagged Meteorological Variables: An Explainable AI Approach," 4th International Conference on Computing Advancements (ICCA 2026), accepted 2026

Tipto Ghosh, **Prottoy Roy**, Abrar Adnan Pranjal, Rubaiyat Islam, Kamruddin Nur, "FloodNet-VQA: A Joint Segmentation and Question Answering Framework for Disaster Assessment," 2026 IEEE International Conference on Signal Processing, Information, Communication and Systems (SPICSCON), BAUET, Natore, Bangladesh, accepted and presented 2026

PROJECTS

Hand Gesture Controller
Computer Vision Project, Python (OpenCV, MediaPipe) github.com/prottoy94/Handtracker_CV

- Built a real-time webcam controller that normalizes 21 hand landmarks and recognizes pinch, fist, and open-hand gestures
- Trained a KNN classifier (scikit-learn) on self-collected data and filtered unstable predictions, so confirmed gestures change slides and mute audio

QuizForge: Online Quiz and Examination Platform
Web Application Project, PHP and MySQL

- Built a role-based platform where instructors publish MCQ quizzes, students take timed attempts and see answer-by-answer results, and scores feed a leaderboard

AIUB Campus Scene Simulation
Course Project, Computer Graphics (C++, OpenGL/GLUT)

- Built an animated campus scene with day and night lighting, moving vehicles and players, seasonal and rainy modes, and particle effects

Other Projects

Donation Management System (Java Swing, Oracle SQL); Zoo Management System (C#); Space Shooter and Ping-Pong Games (C#)

DECLARATION

I hereby declare that all the details furnished above are true to the best of my knowledge and belief.